6ELEN018W Applied Robotics Mock In-Class Test 2025–2026
Duration: 1 Hour 30 minutes

Formatting of this document is not important as the actual test will take place in Blackboard..

Learning outcomes: LO1, LO3, LO4.

Question 1

Which of the following is a valid homogeneous transformation matrix corresponding to a rotation followed by a translation in the 3-D space?

Justify fully your answers by including in your explanations why each of the other choices is not valid. For the valid transformation: state what is the rotation angle, what is the axis we are rotating about and what are the values of the translation components.

A:

$$\begin{pmatrix} 0 & 1 & 0 & t_1 \\ cos(\pi) & 0 & -sin(\pi) & t_2 \\ sin(\pi) & 0 & cos(\pi) & t_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

B:

$$\begin{pmatrix} cos(\frac{\pi}{2}) & 0 & sin(\frac{\pi}{2}) & 0 \\ 0 & 1 & 0 & t_2 \\ -sin(\frac{\pi}{2}) & 0 & cos(\frac{\pi}{2}) & t_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

C:

$$\begin{pmatrix} cos(\frac{\pi}{2}) & 0 & sin(\frac{\pi}{2}) & t_1 \\ 0 & 1 & 0 & t_2 \\ -sin(\frac{\pi}{2}) & 0 & cos(\pi) & t_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} cos(\pi) & 0 & sin(\pi) & t_1 \\ 0 & 1 & 0 & t_2 \\ -sin(\pi) & 0 & cos(\pi) & t_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

D:

$$\begin{pmatrix} cos(\frac{\pi}{2}) & 0 & sin(\frac{\pi}{2}) & t_1 \\ 0 & 1 & 0 & t_2 \\ -sin(\frac{\pi}{2}) & 0 & cos(\frac{\pi}{2}) & t_3 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

E:

$$\begin{pmatrix} cos(\pi) & 0 & -sin(\pi) & t_1 \\ sin(\pi) & 0 & cos(\pi) & 0 \\ 0 & 0 & 1 & t_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(10 marks)

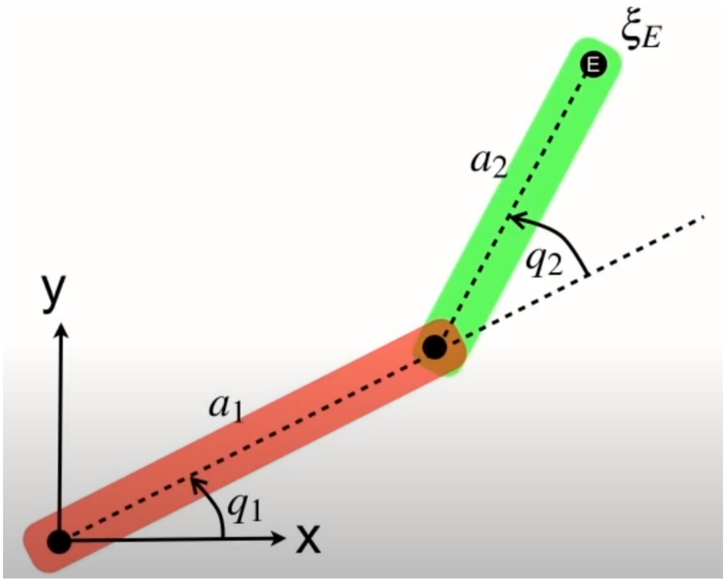
Sample answer:

1 marks for the correct choice of the valid transformation. 1 mark for correct translation angle, axis and translational components. 2 marks for each correct explanation for the other 4 invalid choices.

Option A is not valid because the first row should have 1, 0, 0 and also the cos, sin elements should be in the second and third columns, in order to be valid. Option B is valid rotation about the y-axis with translation components $(0, t_1, t_2)$. Option C is not valid because the rotation angles are not the same. Option D is invalid because the last row should be (0, 0, 0, 1). Option E is invalid because the cos, sin elements should be in the first and second columns.

Question 2

Consider the 2-joint robot in the 2D-space shown below.



1. Describe the series of basic transformations needed to calculate the velocity of the end-effector of the robot ξ_E using the following notation:

   $R(\theta)$ describes the rotation matrix about angle θ, `T_x(a)` describes a translation along the x axis by a units and `T_y(a)` describes a translation along the y axis by a units. *(5 marks)*

2. Without calculating the overall result (i.e. without making the actual matrix multiplications), write down the full form of each homogeneous matrix corresponding to the basic transformations needed above, using variables q_1, q_2, a_1, a_2 and trigonometric functions. *(5 marks)*

3. Describe briefly and include the corresponding formula (without calculating the overall derivatives), how the velocity of the end-effector of the robot can be calculated. *(5 marks)*

4. Write some Python code which calculates and displays the position (x, y) of the end-effector of the 2-joint robot in this Question for the values $q_1 = \frac{\pi}{2}, q_2 = -\frac{2\pi}{3}, a_1 = 5, a_2 = 3$. *(10 marks)*

Correct answer:

1.
$$E = \boldsymbol{R}(q_1) \cdot \boldsymbol{T}_x(a_1) \cdot \boldsymbol{R}(q_2) \cdot \boldsymbol{T}_x(a_2)$$

2.
$$E = \begin{pmatrix} cos(q_1) & -sin(q_1) & 0 \\ sin(q_1) & cos(q_1) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & a_1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} cos(q_2) & -sin(q_2) & 0 \\ sin(q_2) & cos(q_2) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & a_2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

3. The above E transformation matrix contains elements in the last column depending on the q_1, q_2, a_1, a_2 parameters corresponding to $x, y, 1$. The student is not expected to do the actual calculation but to explain that after the calculation E is known. Then:

   From these we can calculate the velocity of the end-effector:

$$\begin{pmatrix} v_x \\ v_y \end{pmatrix} = \frac{\mathrm{d}}{\mathrm{d}t} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix}$$

4. 10 marks for a completely correct solution with correct syntax too. 6-9 marks for minor mistakes in implementation and syntax. 3-5 marks for an attempt which is almost half correct or so. 0-2 marks for a very poor totally incorrect attempt.

```python
import numpy as np
from math import *

q1 = pi/2
q2 = -2*pi/3
a1 = 5
a2 = 3

R_q1 = np.array([[cos(q1), -sin(q1), 0],
       [sin(q1), cos(q1), 0],
       [0,         0,     1]])

R_q2 = np.array([[cos(q2), -sin(q2), 0],
       [sin(q2), cos(q2), 0],
       [0,         0,     1]])

T_xa1 = np.array([[1, 0, a1],
         [0, 1, 0],
         [0, 0, 1]])

T_xa2 = np.array([[1, 0, a2],
         [0, 1, 0],
         [0, 0, 1]])


T = R_q1@T_xa1@R_q2@T_xa2

print('x=', T[0,2], 'y=', T[1,2])
```

Question 3

1. Describe briefly what is the Jacobian matrix. Give the general formula for calculating the velocity of the end-effector of a robot, given the Jacobian matrix and the angular velocities of the robot joints. *(4 marks)*

2. You are required to control the end-effector velocity of a robot. Given the Jacobian matrix for the specific mechanics of the robot how you would calculate the angular velocities that the joint actuators need to produce? Give the formula for the calculation of the required desired joint velocities. *(5 marks)*

Sample answer:

1. The Jacobian can be considered as the derivative of a matrix vector. In robotics and for this problem it connects the velocity of the end-effector of a robot with the angular velocities of the joints of the robot.

   To calculate:

   $$\boldsymbol{v} = \boldsymbol{J}(\boldsymbol{q})\dot{\boldsymbol{q}} \tag{1}$$

   where $\boldsymbol{J}(\boldsymbol{q})$ is the Jacobian of the joint angles q_1, q_2

2. As we know the end-effector velocity we can use the same formula from the previous subquestion and solve for the required joints angular velocities. From matrix algebra this can be done using:

   $$\dot{\boldsymbol{q}} = \boldsymbol{J}(\boldsymbol{q})^{-1} \cdot \boldsymbol{v} \tag{2}$$

   where $\boldsymbol{J}(\boldsymbol{q})^{-1}$ is the inverse of the Jacobian matrix.

Question 4

Consider a robot which tries to hold a rectangular block object (rigid body) with its 2 fingers. The weight force W is applied at the centre of mass. The 2 fingers apply 2 forces F_1 and F_2 at the top and the bottom of the object respectively and these forces are at a distance x as shown in Figure 1.

1. Describe the necessary condition(s) so that the rectangular rigid body block reaches equilibrium (it does not move). Give the formulas required in order to achieve this, based on the problem parameters shown in the figure. *(6 marks)*

2. Derive the equations for forces F_1, F_2 which are required to achieve the rigid body equilibrium. Show all the steps of your derivation. *(9 marks)*

Sample answer:

Marks awarded: 3 + 3 for each formula.

9 marks for full solution of the system. 5-8 for most of the derivation being OK. 2-4 for just a basic start of the derivation. 0-1 marks for no or simply completely incorrect/irrelevant solution.

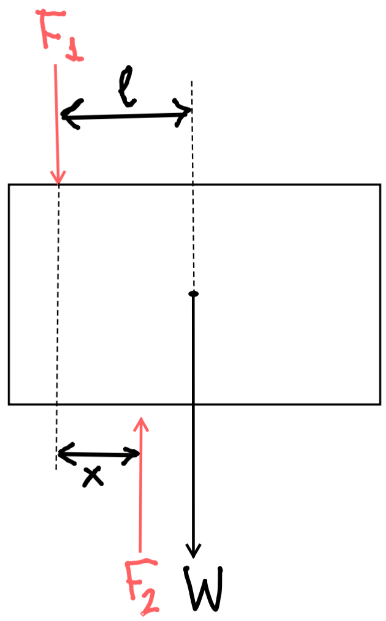


Figure 1: The robot gripping problem of Question 4.

1. Apply the equilibrium principles for a rigid body, balance both forces and torques:

   To balance the forces:

   $$F_2 = F_1 + W$$

   To balance the torques:

   $$F_1 * l - F_2 * (l - x) = 0$$

2. Solve this system of equations for the forces of the 2 robot fingers.

Solution:

$$F_1 + W - F_2 = 0 \Rightarrow F_1 = F_2 - W$$

$$F_1 * l - F_2 * (l - x) = 0 \quad (2)$$

$$(2) \xrightarrow{(1)} (F_2 - W) * l - F_2 * (l - x) = 0 \Rightarrow$$

$$F_2 * l - W * l - F_2 * l + F_2 * x = 0$$

$$\Rightarrow$$

$$\cancel{F_2 * l} - W * l - \cancel{F_2 * l} + F_2 * x = 0$$

$$\Rightarrow$$

$$-W * l + F_2 * x = 0 \Rightarrow$$

$$\boxed{\begin{aligned} F_2 &= \frac{W * l}{x} \quad (3) \\ \xrightarrow{(1)} F_1 &= \frac{W * l}{x} - W \quad (4) \end{aligned}}$$

Question 5

A point P with coordinates $[8, 2, 15]^T$ is attached to the end-effector of a robot, a moving frame F, which is subject to the following 3 successive transformations relative to the reference fixed world frame F_{xyz}:

1. Rotation of 60^o about the z-axis.
2. Followed by a rotation of 60^o about the y-axis.
3. Followed by a translation of $[5,\ -2,\ 3]$.

Write some Python code which calculates the coordinates of the point relative to the reference frame after all the transformations have taken place.

(15 marks)

Award of marks: correct transformations chosen: 5 marks. correct order of application of transformations: 4 marks. Python code correct implementation of the previous: 6 points. Python syntax almost totally incorrect: -3 marks

```
import numpy as np
from math import *

P = [8, 2, 15, 1]  # 1 added to make calculations valid

# angle 60 degrees
a = radians(60)
tr1 = [[cos(a), -sin(a), 0, 0],
       [sin(a), cos(a), 0, 0],
       [0,         0,    1,  0],
       [0, 0, 0, 1]]

tr2 = [[cos(a), 0, sin(a), 0],
       [0,      1,  0,     0],
       [-sin(a), 0, cos(a), 0],
       [0, 0, 0, 1]]

tr3 = [[1, 0, 0, 5],
       [0, 1, 0, -2],
       [0, 0, 1, 3],
       [0, 0, 0, 1]]

P = np.array(P)
tr1 = np.array(tr1)
tr2 = np.array(tr2)
tr3 = np.array(tr3)

result = tr3@tr2@tr1@P
print(result)
```

Question 6

1. Briefly describe the Euler-Lagrange method to calculate the dynamic equations of motion for a robot. *(5 marks)*

2. What are the advantages of the Euler-Lagrange method compared with the Newtonian mechanics method? *(4 marks)*

Sample answer:

1. Lagrangian is the difference between kinetic and potential energies of the system.

$$\mathcal{L} = \mathcal{K} - \mathcal{P} \tag{3}$$

   To calculate the dynamic equations of motion for a system such as a serial-link robot according to the Euler Lagrange method: write the kinetic and potential energies of the system in terms of a set of *generalised coordinates* $(q_1, q_2, \ldots q_n)$ where n is the degrees of freedom of the system. q_k can be linear distances or angles:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_k} - \frac{\partial \mathcal{L}}{\partial q_k} = \tau_k, \quad k = 1, \ldots, n$$

   where τ_k is the (generalised) force (linear force or torque) associated with q_k

2. • Easier for more complicated systems.
   • Based on system's energies
   • Systematic approach

Question 7

Describe briefly 3 different ethical and social issues related to robotics. Justify your answer for each of these 3 issues.

(9 marks)

Correct answer:

There is a number of different correct answers. 2 marks per ethical/social issue for a valid answer and 1 mark per ethical/social issue for proper justification.

Question 8

Consider a reference frame A fixed to the robot's end-effector and a reference frame B which is the world frame. Assuming that we know the matrix for the pose of B relative to A, how can we determine the pose of A relative to B ?

Implement a Python function for your answer given above. The function accepts as a single argument a list corresponding to the pose of B relative to A. It returns a list or an array corresponding to the pose of A relative to B.

(8 marks)

Sample answer:

Correct determination of pose 3 marks. Correct implementation in Python 3 marks. Python function syntax correctness: 2 marks.

The inverse transformation needs to be calculated based on transformation matrix known.

```
import numpy as np

def calculate_pose(pose):
   A = np.array(pose)
   new_pose = np.linalg.inv(A)

   return new_pose
```